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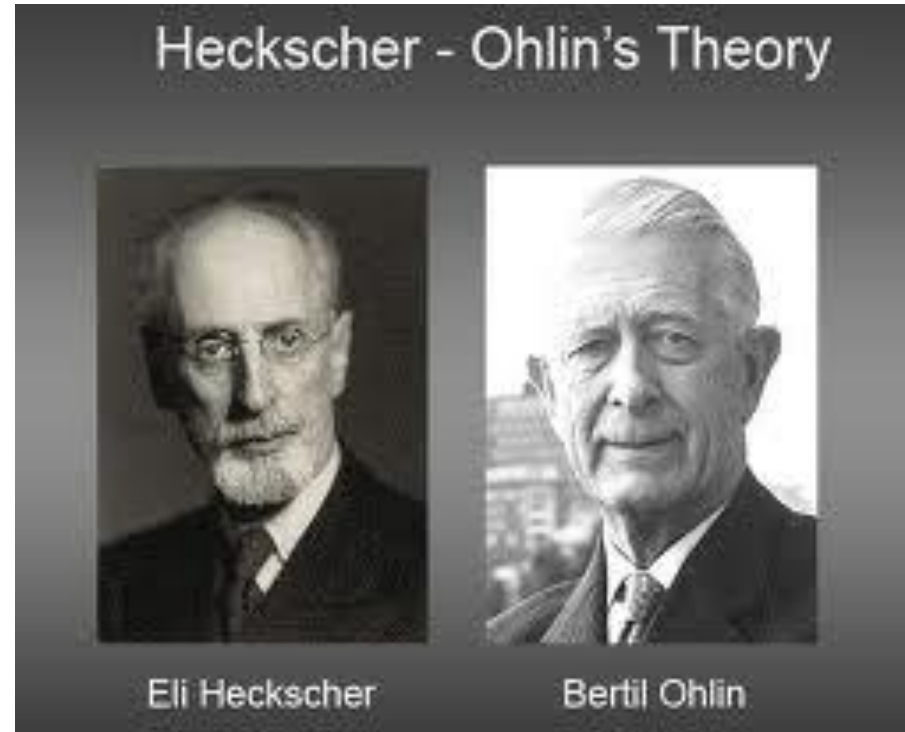
Heckscher-Ohlin Theory of Trade

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- Extending trade model to include:
 - Basis of comparative advantage
 - Effect of international trade on return to labor



Assumptions of the Theory

- **Heckscher-Ohlin theory** based on following assumptions:
 1. Two nations, two goods, two factors of production
 2. Technology same in both nations
 3. Commodity X is labor intensive, commodity Y is capital intensive in both nations
 4. Constant returns to scale for X and Y in both nations
 5. Incomplete specialization in production in both nations



Other Assumptions

- **Heckscher-Ohlin theory** based on following other assumptions:
 6. Tastes are equal in both nations
 7. Both commodities and factors are traded in perfectly competitive markets
 8. Perfect factor mobility within each nation, but not between nations
 9. No transportation costs, tariffs or other barriers to free trade.
 10. All resources are fully employed in both nations
 11. International trade between the nations is balanced.



- Perfectly mobile factors within each country
 - Same wage rate earned in both industries of each country
 - Same rental rate earned in both industries of each country
- Identical and homothetic preferences across countries
 - A poorer country will buy less of both rice and airplane, but in the same ratio as a richer country facing the same prices
 - Not a realistic assumption, but it allows us to focus attention on the differences in resources as the sole reason for trade

Factor Intensity, Factor Abundance

- **Factor Intensity**

- In a two-commodity, two factor world, commodity Y is ***capital intensive*** if the capital-labor ratio (K/L) used in the production of Y is greater than K/L used in the production of X.
- It is not the absolute amount of capital and labor used in production of X and Y, but the amount of *capital per unit of labor* that determines capital intensity.



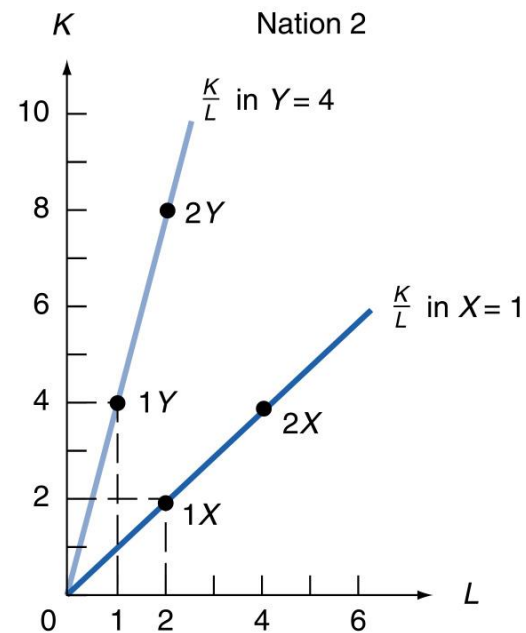
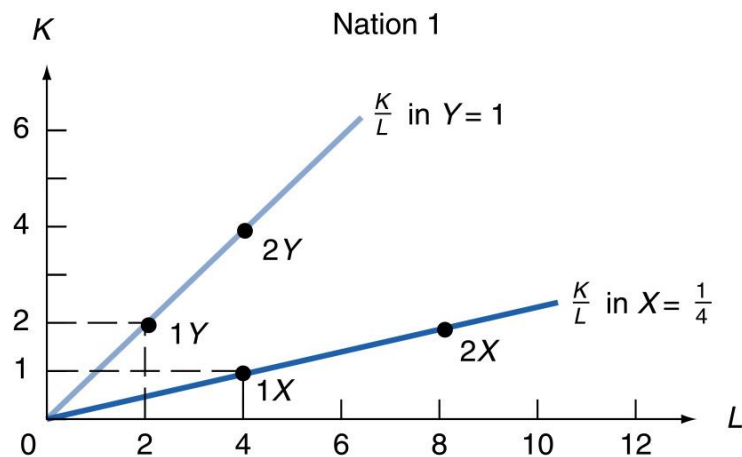


FIGURE Factor Intensities for Commodities X and Y in Nations 1 and 2.

Factor Intensity, Factor Abundance

- **Factor Abundance**

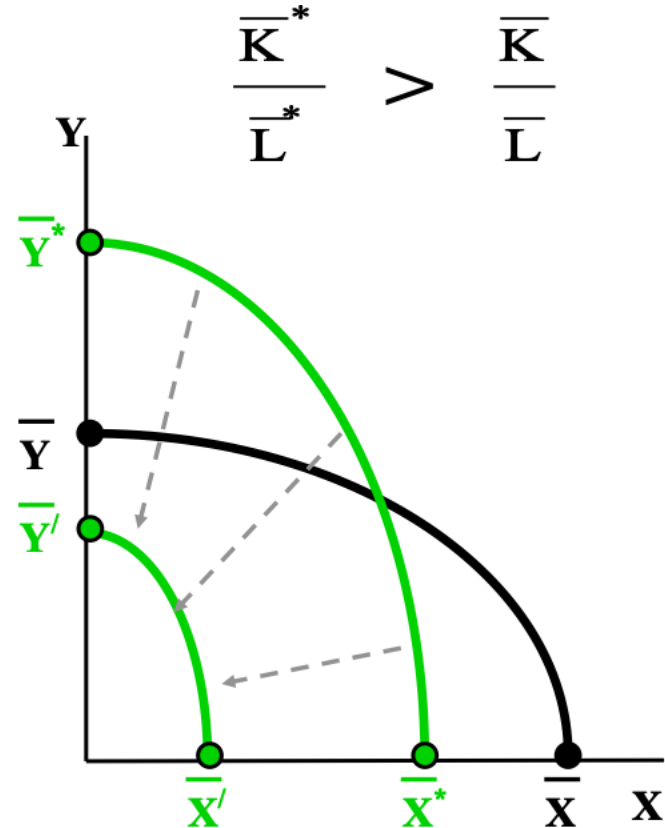
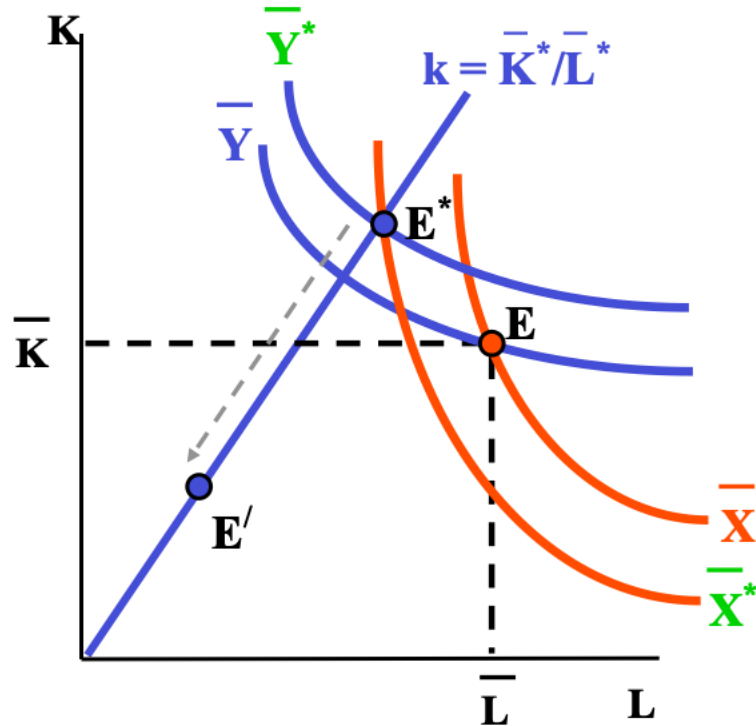
- In terms of *physical units*:

$$\frac{\bar{K}^*}{\bar{L}^*} > \frac{\bar{K}}{\bar{L}}$$

- Nation 2 is capital abundant if the ratio of the total amount of capital to the total amount of labor (TK/TL) available in Nation 2 is greater than that in Nation 1.
- It is not the absolute amount of capital and labor available in each nation, but the *ratio* of the total amount of capital to the total amount of labor.



Factor Abundance and PPCs



Factor Intensity, Factor Abundance, and the Shape of the Production Frontier

- **Factor Abundance**

- In terms of *relative factor prices*:

- Nation 2 is capital abundant if the ratio of the rental price of capital to the price of labor time (P_K/P_L) is *lower* in Nation 2 than in Nation 1.
 - Rental price of capital is usually considered to be the interest rate (r), while the price of labor time is the wage rate (w), so $P_K/P_L = r/w$.
 - It is not the absolute level of r that determines whether a nation is K-abundant, but r/w .



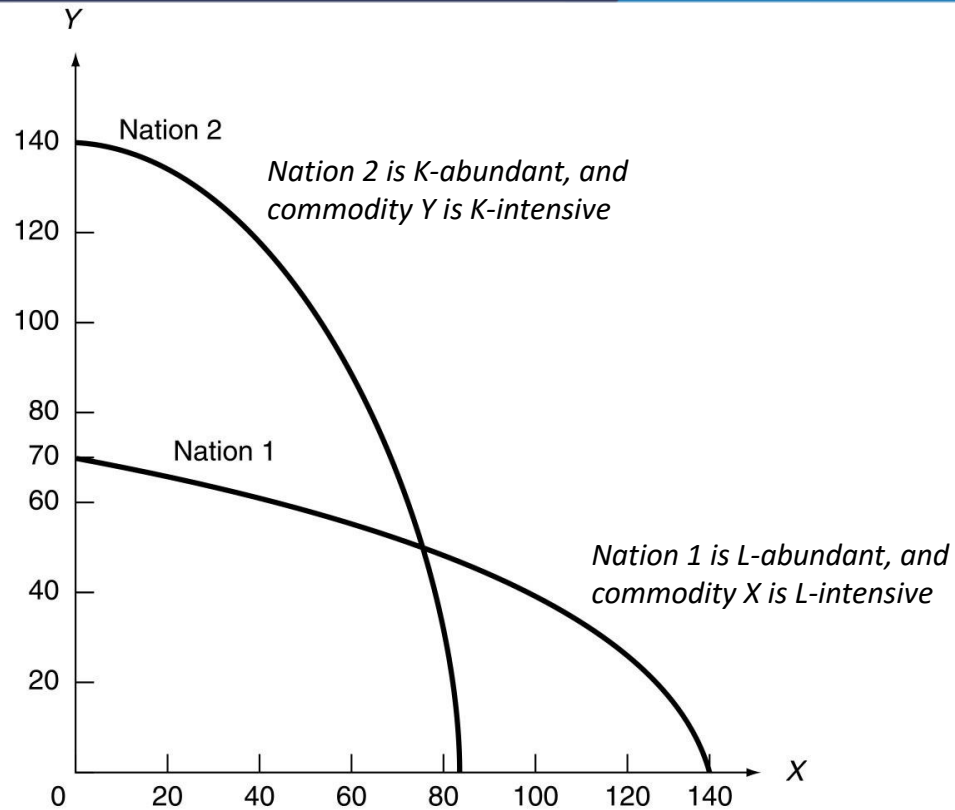
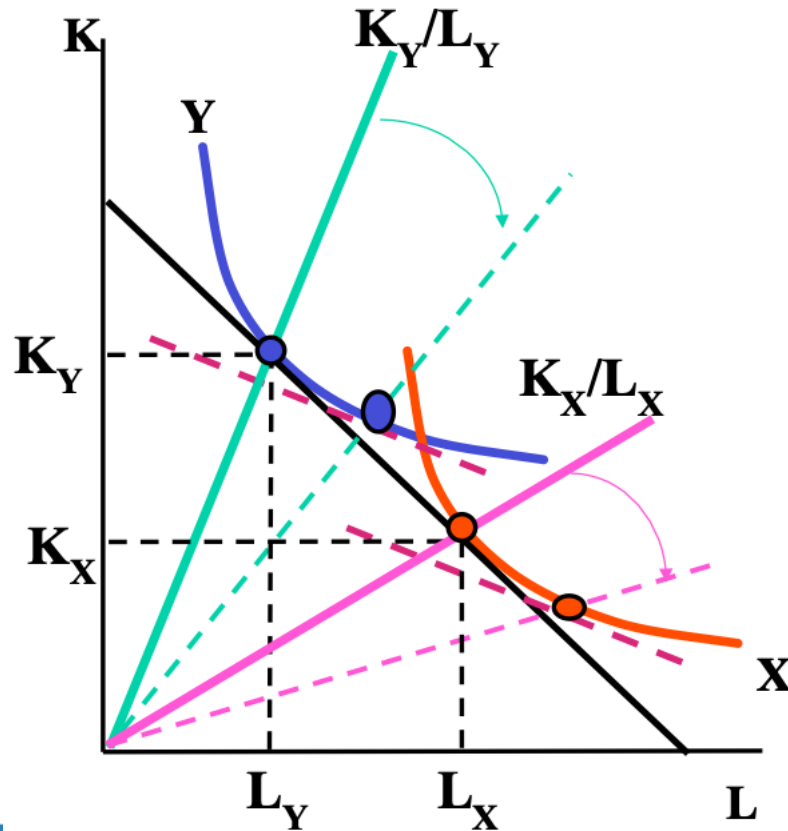


FIGURE The Shape of the Production Frontiers of Nation 1 and Nation 2.

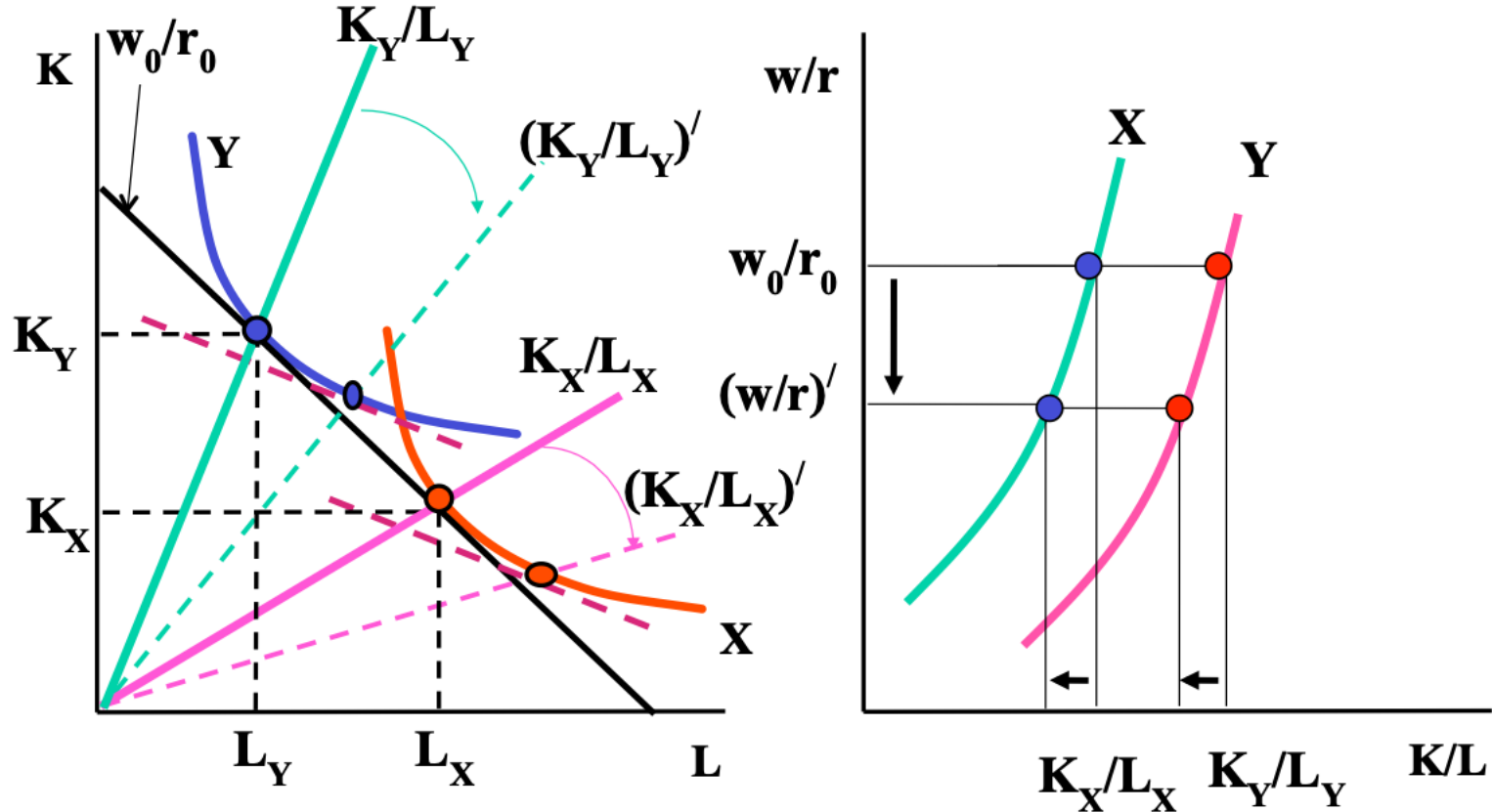
No Factor Intensity Reversal



- Factor intensity is the relative capital-labor ratio. It refers to the demand for factors of the goods.
- Regardless of the changes in w/r , the factor intensity of one good relative to that of other goods does not change.
- It is assumed here that Y is always capital intensive while X is labor intensive or the capital to labor demand of Y is always greater than those of X.

$$\frac{K_Y}{L_Y} > \frac{K_X}{L_X}$$

Factor Intensity and Capital-labour Ratio



Factor Endowments and the Heckscher-Ohlin Theory

- **Heckscher-Ohlin (H-O) theory** is based on two theorems:

1. The H-O theorem

- *A nation will export the commodity whose production requires the intensive use of the nation's relatively abundant and cheap factor and import the commodity whose production requires the intensive use of the nation's relatively scarce and expensive factor.*



1. The H-O theorem

- In short, the relatively labor-rich nation exports the relatively labor-intensive commodity and imports the relatively capital-intensive commodity
- *Explains* comparative advantage rather than assuming it.

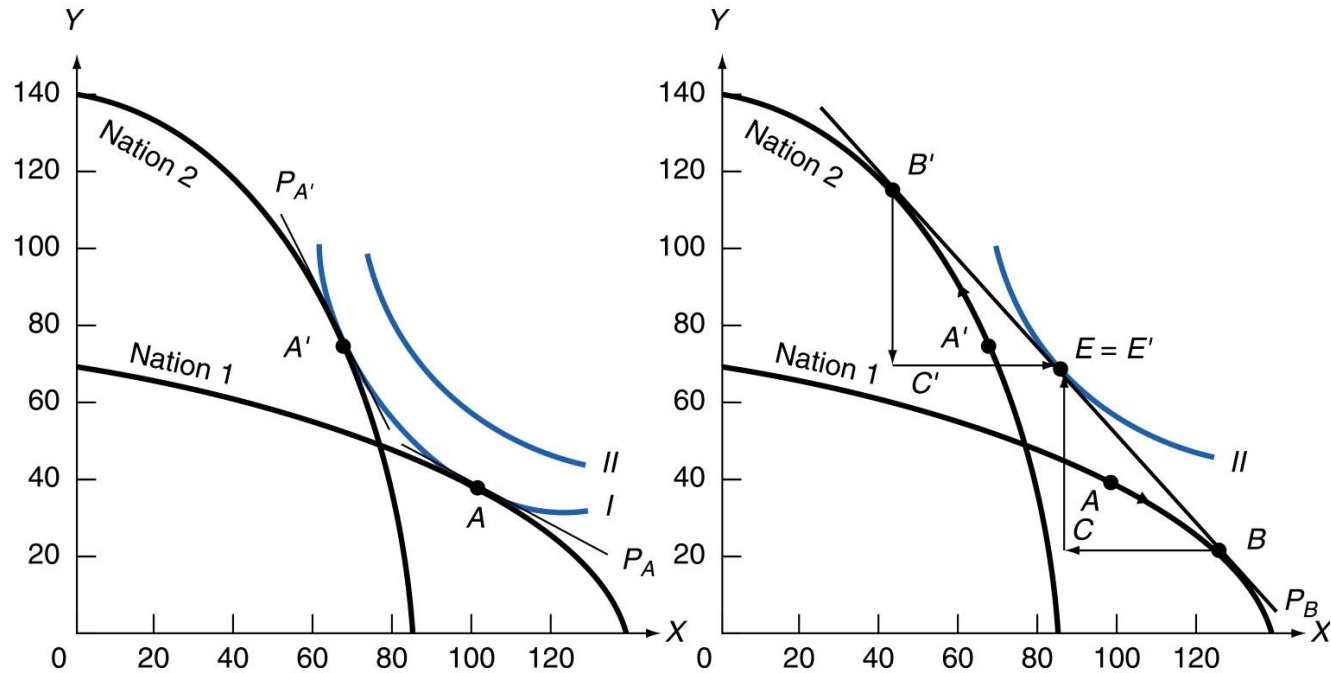


FIGURE The Heckscher-Ohlin Model.

Factor-Price Equalization and Income Distribution

- **Heckscher-Ohlin (H-O) theory** is based on two theorems:

2. The factor price equalization theorem

- *International trade will bring about equalization in the relative and absolute returns to homogenous factors across nations.*
- In short, wages and other factor returns will be the same after specialization and trade has occurred.
- Holds only if H-O theorem holds.



Factor-Price Equalization and Income Distribution

2. The factor price equalization theorem

- International trade causes w to rise in Nation 1 (the low-wage nation) and fall in Nation 2. (the high-wage nation), reducing the pre-trade difference in w between nations.
- Similarly, trade causes r to fall in Nation 1 (the K-expensive nation) and rise in Nation 2. (the K-cheap nation), reducing the pre-trade difference in r between nations.

Factor-Price Equalization and Income Distribution (across Nations)

- H-O-S Theory: after Samuelson received Nobel prize in 1970

2. The factor price equalization theorem

- International trade brings about equalization in the relative and absolute returns to homogenous factors across nations.
- Thus, international trade causes a redistribution of income from the relatively expensive (scarce) factor to the relatively cheap (abundant) factor.



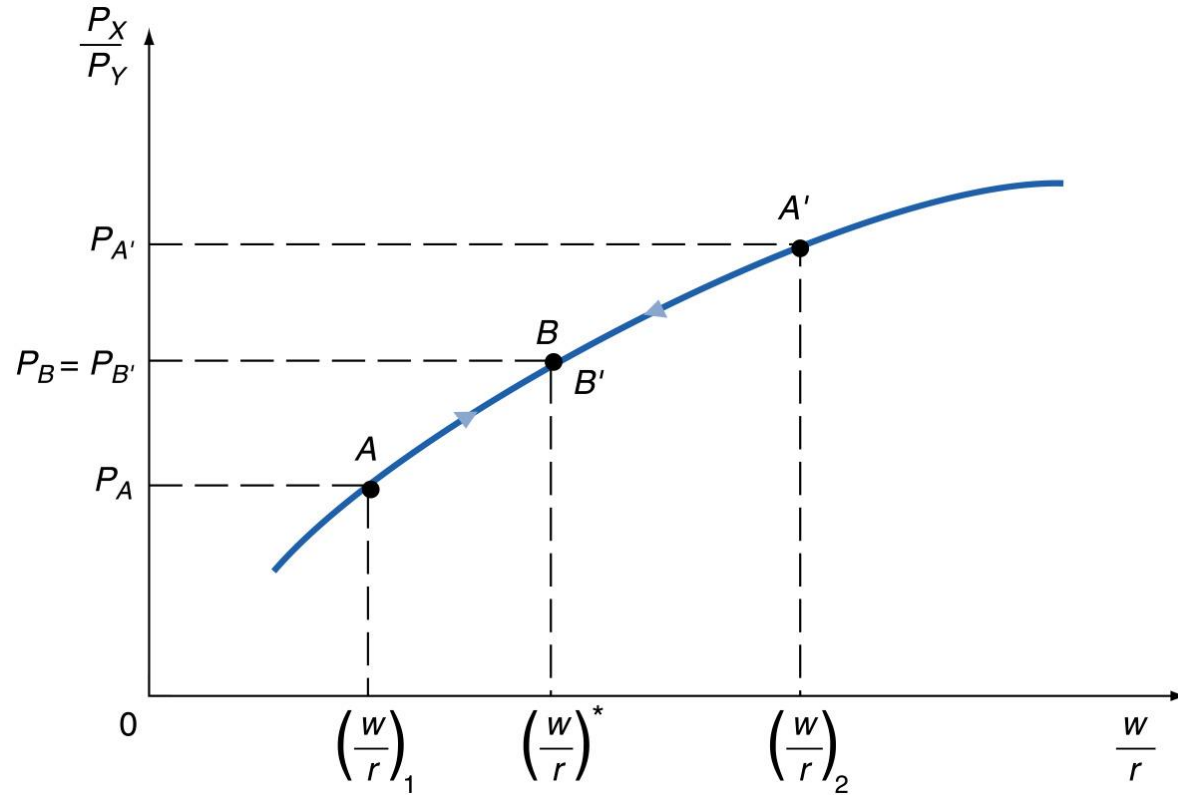


FIGURE Relative Factor–Price Equalization.

Questions

Q1. A nation is said to have a relative abundance of K if it has a:

- a. greater absolute amount of K
- b. smaller absolute amount of L
- c. higher L/K ratio
- d. lower r/w

Ans: d

- Q2. Factor Price Equalization means that,
- a. All workers are equally productive.
 - b. If a country fails to trade, its skilled workers will earn no more than its unskilled workers.
 - c. Trade causes the return to human capital to be the same as the return to physical capital.
 - d. For countries to trade freely, they must tax factors of production so that firms in all countries pay the same factor prices.
 - e. Free trade causes identical factors in different countries to be paid more nearly the same than they were in autarky.

Ans: e

*Thank
you*

